

Community Library Membership & Book Lending System

→ before making entities and DbContext in Code, I designed the Database.

Admin Table:

- Id (PK)
- Name
- PhoneNum
- Password
- IsActive
- CreatedAt
- UpdatedAt

Members Table:

- Id (PK)
- Name
- Email (unique)
- PhoneNum
- Password
- MembershipType (enum: Basic, Premium, Student)
- MembershipStatus (enum: Active, Inactive)
- IsActive
- AllowedBorrowingCount
- CurrentBorrowedCount

- CreatedAt
- UpdatedAt

BookCategory Table:

- Id (PK)
- CategoryName (unique)
- IsActive
- CreatedAt
- UpdatedAt

Books Table:

- Id (PK)
- CategoryId (FK to BookCategory table)
- Title
- Author
- ISBN (unique)
- PublicationYear
- NoOfCopies
- IsActive
- CreatedAt
- UpdatedAt

BookCopies Table:

- Id (PK)
- BookId (FK to Books table)
- BookCopyId (unique)
- IsAvailable
- IsBorrowed
- IsDamaged
- ConditionNotes
- CreatedAt
- UpdatedAt

Borrowings Table:

- Id (PK)
- MemberId (FK to Members table)
- BookCopyId (FK to BookCopies table)
- BookId (FK to Books table)
- BorrowedDate
- DueDate
- MemberReturnedDate
- FineAmount
- Status (enum: Borrowed, Returned, Overdue)
- IsActive

- CreatedAt
- UpdatedAt

BorrowingRules Table:

- Id (PK)
- MembershipType (enum: Basic, Premium, Student)
- MaxActiveBorrowings
- MaxBorrowDays
- CreatedAt
- UpdatedAt

Fines Table:

- Id (PK)
- BorrowingId (FK to Borrowings table)
- MemberId (FK to Members table)
- FineAmount
- FineReason
- PaymentStatus (enum: Paid, Pending)
- PaymentDate
- CreatedAt
- UpdatedAt

Relationships:

- Books table has a many-to-one relationship with BookCategory (one category has many books)
- BookCopies table has a many-to-one relationship with Books (one book has many copies)
- Borrowings table has many-to-one relationships with Members, BookCopies, and Books
- Fines table has many-to-one relationships with Borrowings and Members
- BorrowingRules table is independent with MembershipType as the key differentiator